

1. Executive Overview & The \$2.9B BEC Fraud Crisis

Every year, enterprises lose over **\$2.9 billion** to Business Email Compromise (BEC) and vendor impersonation fraud. Traditional security tools rely on spam filters or text anomaly classifiers. When an attacker registers a lookalike domain (e.g., acme-payments.co vs official acme.in) or compromises an executive mailbox with authentic PDF invoices, spam filters fail completely.

■ The Core Architectural Distinction:

VerifyFlow is a pre-action verification agent, not a phishing classifier. It does not attempt to guess whether an email text 'feels' malicious. Instead, it is a bounded decision-making agent that autonomously determines what must be independently verified across authoritative ledgers before an irreversible, high-consequence financial transaction is permitted to execute.

100%

BENCHMARK ACCURACY

0 / 29 (0.0%)

UNSAFE APPROVAL RATE

-28.5%

TOOL CALL REDUCTION

89 / 89

REGRESSION TESTS GREEN

2. Dynamic Agent Investigation: The \$125,000 BEC Wire Attack

VerifyFlow executes an iterative **Observe → Decide → Act → Evaluate (OODA)** cognitive cycle. Below is the live execution path for an urgent \$125,000 invoice claim from 'Acme Supplies':

Live Agent Reasoning Steps (Case B):

- **1. OBSERVE:** Extracts invoice claims (\$125,000, INV-4938, urgent terms).
- **2. VENDOR BASELINE:** Queries vendor registry → *Acme Supplies is trusted (18 past payments)*.
- **3. DOMAIN AUDIT:** Queries domain authority → **CONFLICT:** *acme-payments.co ≠ official acme.in.*
- **4. BANK AUDIT:** Evaluates destination account → **CONFLICT:** *BANK-ACME-999 differs from trusted account.*
- **5. INDEPENDENT VERIFICATION:** Attempts out-of-band verification to resolve ambiguity.

Authoritative Policy Verdict:

■■ **BLOCKED — HELD FOR REVIEW**

- **Deterministic Risk Score:** 100 / 100 (CRITICAL)
- **Signals:** Lookalike domain (+30) + Bank change (+40) + Unverified (+20)
- **Capital Protected:** \$125,000.00 wired funds prevented.
- **Safety Guarantee:** No black-box LLM hallucination can authorize funds.

3. Multi-Step Runtime Failure Adaptation & Self-Healing AI

In production, external APIs go down. VerifyFlow treats **intermediate tool failure as an actionable state transition** rather than an unhandled error:

1. Injected Tool Outage

Primary verification API returns TOOL_UNAVAILABLE (Simulated portal outage).

2. Autonomous Replanning

Agent enters ADAPTING state and dynamically invokes secondary contact line (verify_via_trusted_contact).

3. Productive Recovery / Lockdown

- CFO confirms change → **APPROVE** (Self-healed).
- CFO repudiates → **FRAUD QUARANTINE.**

4. Quantitative Evaluation: The 52-Scenario Controlled Benchmark

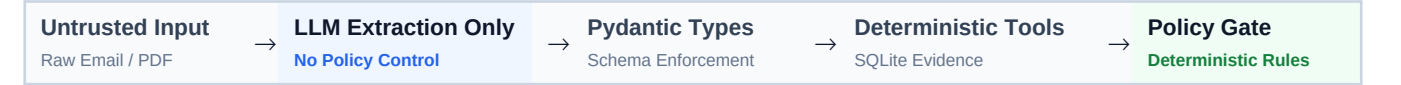
To rigorously validate VerifyFlow, we authored an independent benchmark of **52 controlled operational scenarios** spanning 5 comprehensive architectural categories. All expected results were defined prior to evaluation runs:

Scenario Category	Cases	Accuracy	Unsafe Approvals	Key Operational Focus
1. Clean & Routine Invoices	10	100% (10/10)	0	Zero friction on verified suppliers (\$1k to \$100k)
2. BEC Attacks & Lookalike Domains	14	100% (14/14)	0	Typosquatting, unverified bank swaps, urgency coercion
3. Outage & Multi-Step Adaptation	12	100% (12/12)	0	Primary tool failures with secondary self-healing recoveries
4. Adversarial Edge Cases & Boundaries	8	100% (8/8)	0	Prompt injections ('ignore previous'), \$0 bounds, rush orders
5. Invariant Stress & Cycle Limits	8	100% (8/8)	0	Tool infinite loops, cycle capping, randomized invariant fuzzing
TOTAL EMPIRICAL BENCHMARK	52	100% (52/52)	0 / 29 (0.00%)	Zero critical false approvals across entire suite

5. Comparative Baseline: Why Agentic Verification Wins

Architecture	Accuracy	Unsafe Approvals	Avg Tools	Operational Bottleneck / Failure Mode
Static 6-Tool Rule Pipeline	100%	0	6.00 (Fixed)	Brute force: executes every tool on every invoice (High API cost)
One-Shot LLM Classifier	88.5%	6 (CRITICAL FAIL)	0.00	Easily tricked by prompt injection, urgent text, and lookalike domains
VerifyFlow (Adaptive Agent)	100%	0 / 29 (0.00%)	4.29 (-28.5%)	Optimal: Early stopping on clean/rejected cases; self-heals on outage

6. Defense-In-Depth Security Boundaries



■ **Security Guarantee:** The LLM has zero authorization authority. Evidence is written to an immutable SQLite ledger with SHA-256 checksums. Final payment disbursement requires explicit approval by deterministic Python policy rules. Adversarial prompts cannot bypass this gate.

7. Production Access & Deployment Links

Deployment & Codebase Links: • GitHub Repository: https://github.com/Murshid-CSE/agent-ai • GitHub Pages Live URL: https://murshid-cse.github.io/agent-ai/ • Local Command Center: http://localhost:5173/ • FastAPI Swagger Docs: http://127.0.0.1:8000/docs	Verification & Engineering Quality: • Pytest Backend Suite: 89/89 tests passing (100% green) • Frontend Production Build: Vite built in 698ms (0 errors) • Demo Video Studio: Built-in interactive 5-scene player & recorder • Resilience Mode: Full offline execution capability
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